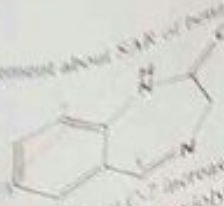
C. Calcium
D. Potassium

2. Identify correct statement about SAR of benzodiazepines

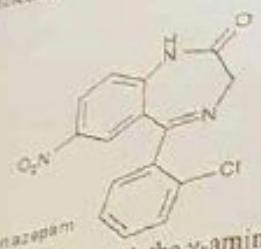


A. Chlorine substitution at C-7 increase the activity
B. Substitution at C-10 increase anxiolytic activity
C. Replacing C-2 oxygen with sulfur decrease anxiolytic activity
D. Substitution of phenyl group at C-5 decrease activity

3. Which of the following amino acid serve as hydrogen source during interaction of proton acceptor group present at ring-B of benzodiazepines with GABA receptors

A. Histidine
B. Lysine
C. Aspartate
D. Asparagine

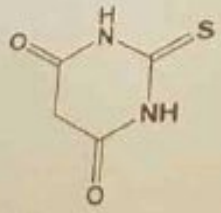
4. Benzodiazepines, exemplified here by clonazepam, produce their antiseizure effects by which of the following mechanisms?



Clonazepam

A. Inhibiting the influx of chloride through the γ-aminobutyric acid B ($GABA_B$) channel by binding to an allosteric site.
B. Directly binding to the voltage-gated sodium channel and inhibiting the influx of sodium.
C. Enhancing the influx of sodium through its voltage-gated channel by binding to an allosteric
D. Enhancing the effect of γ-aminobutyric acid (GABA) on the $GABA_A$ chloride channel by allosterically increasing the affinity of GABA for its receptor.
E. Directly binding to the γ-aminobutyric acid A ($GABA_A$) channel and acting as a GABA mimetic agent.

5. Which of the following is true about the barbiturate analogue shown below?



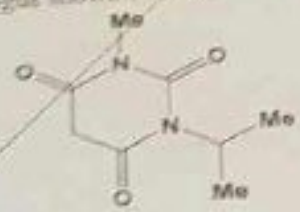
A. Is less polar than Amobarbital
B. Is inactive as a barbiturate
C. Enhances GABAergic inhibitory response in the CNS

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Time allowed: - 3 hours

6. From the Structural Activity Relationship studies of barbiturates, what can be inferred about the activity of the analogue shown below?



A. has an Improved acidity than Barbituric acid
B. Is more basic than Barbituric acid
C. Is equipotent with Barbituric acid
D. Is a 5, 5-disubstituted analogue

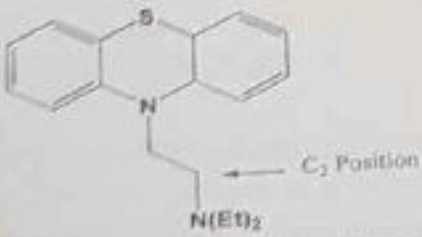
7. All are true about the SAR of alcohol and their derivatives as sedative-hypnotics **except**?

A. Branching of the alkyl chain improves activity
B. Indefinitely increasing chain length improves activity
C. Introduction of a halogen in the alkyl chain improves activity
D. Carbamylation increases depressant potency

8. Phenytoin is a drug used for the treatment of generalized tonic-clonic seizure. The ion channel affected most commonly by this drug is

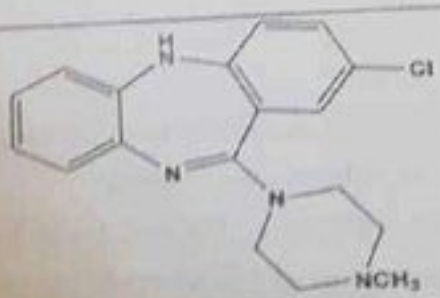
A. Sodium
B. Potassium
C. Calcium
D. Chlorine

9. By considering the SAR of tricyclic tranquilizers, identify <u>incorrect</u> statement about the compound shown below?



A. Has a lower activity compared to chlorpromazine
B. Is useful in the treatment of Parkinson's disease
C. Phenyl substitution at C_2 position improves neuroleptic potency
D. All of the above

10. Which of the following is <u>true</u> about the drug with the structure shown below?

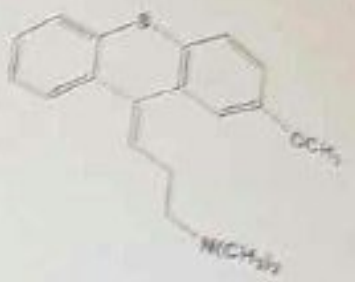


I= is a member of the Dibenzazepines	III= has neuroleptic and anticholinergic propert
II= its generic name is clozapine	IV= does not produce extrapyramidal side effec

A. I only
B. I and II
C. I, II and III
D. All of the above

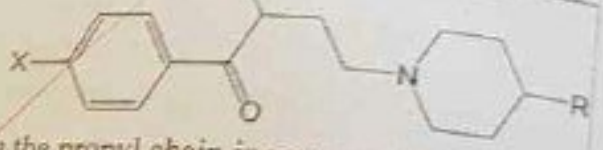
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11. Shown below under is the structural analogue of the tricyclic neuroleptic agent. Which is true regarding this compound?



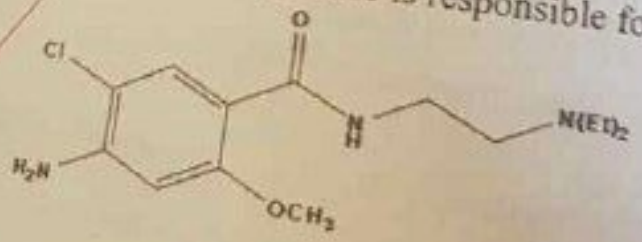
A. The methoxy substitution at position 2 improves neuroleptic activity
B. Can exhibit geometric isomerism
C. Is a member of the dibenzazepine class of neuroleptics
D. All of the above

12. Identify the **correct** statement about the SAR of the following structure



A. Lengthening the propyl chain increase activity of the drug
B. Reduction of carbonyl group increase the potency of the drug
C. Fluorine substitution at the para position of aromatic ring increase potency of the drug
D. None

13. Benzamides are a class of drug used as antipsychotic agent and their antipsychotic effect is from the observation that rodents administered with antiemetic drug metoclopramide exhibit behavioral properties consistence with antagonism of dopamine receptors. Which of functional present on the structure of metoclopramide is responsible for antagonism of the receptors



A. Amino
B. Chlorine
C. Methoxy
D. Ketone

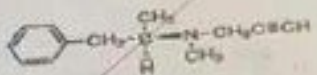
14. What is the benefit of coadmnistering levodopa with carbidopa
A. Levodopa inhibit the decarboxylase enzyme
B. Carbidopa increase concentration of levodopa in the brain
C. Carbidopa incease metabosim of levodopa by decarboxylase enzyme
D. None

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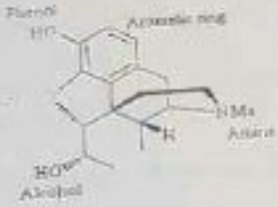
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15. Which one of the following is **true** about the following structure



A. Its MAO inhibitor
B. Its Catechol-O-Methyltransferase enzyme Inhibitors
C. IIS generic name is selegiline
D. A and C

16. The following structure is an important medicine.



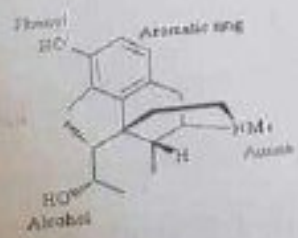
What is its major clinical use?

A. An anti-ulcer agent
B. A cardiovascular agent
C. An analgesic
D. An anti-asthmatic agent

17. Morphine is an important drug to alleviate pain. Which is the most serious long term side effect of morphine?

A. Tolerance and depression
B. Constipation
C. Depression of the respiratory center
D. Pupil constriction

18. Shown here under is the structure of an important analgesic.



What interactions are involved in binding of ether group of morphine to the target binding site?

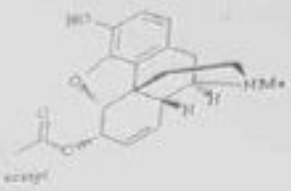
A. Ionic interactions
B. Hydrogen bonding interactions
C. Van der Waals binding interactions
D. None of the above

Name:-
Time allowed: - 3 hours

19. Morphine is an important analgesic. Which of morphine's side effects can be used to advantage another aspect of medicine?

A. Tolerance and depression
B. Diarrhea
C. Depression of the respiratory system
D. Pupil construction

20. The following structure **is more active** than morphine as an analgesic



What is the principle reason for this?

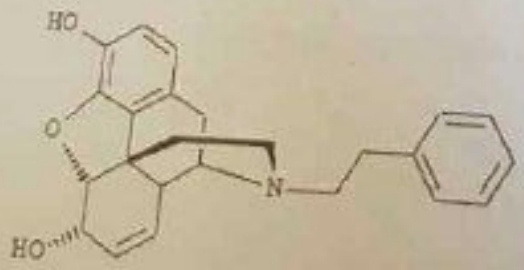
A. The extra acetyl group interacts with an extra binding region within the binding site by acting as a hydrogen bonding acceptor.

B. The extra acetyl group masks a polar group allowing the structure to cross the blood brain barrier to move efficiently.

C. The extra acetyl group masks a polar group allowing the structure to cross cell membranes more efficiently.

D. The extra acetyl group masks an alcohol group which would otherwise undergo a phase II metabolic conjugation reaction leading to rapid excretion of the drug.

21. The *N*-phenethyl analogue of morphine has an analgesic activity which is 14 times greater than that of morphine. What is the principle reason for this?

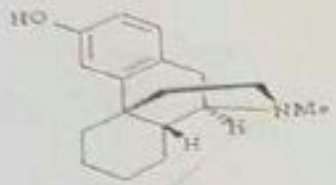


A. the aromatic ring of the phenethyl group interacts with an extra hydrophilic region of the binding site
B. the aromatic ring of the phenethyl group interacts with an extra hydrophobic region of the binding site
C. the phenethyl group increases the hydrophobicity of the molecule allowing a greater percentage to cross the blood brain barrier
D. the phenethyl group acts as a steric shield to protect the amine group from drug metabolism

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22. Levorphanol is an analgesic which is five times more active than morphine.



What drug design strategy has been used in developing this structure from the lead compound morphine?

A. Simplification
B. Rigidification
C. Extension
D. Substituent variation

23. Metazocine has similar analgesic activity to morphine. To which group of compounds does this structure belong?

A. Benzomorphans
B. 4-phenylpiperidines
C. Morphinans
D. Enkephalins

24. Etorphine is used in veterinary medicine. To which class of compounds does it belong?

A. Benzomorphans
B. Endorphins
C. Oripavines
D. Endomorphins

25. What pharmacophore is shared by the opiate analgesics?

A. A charged nitrogen, a phenol and an aromatic ring
B. A charged nitrogen, an alcohol and an aromatic ring
C. A charged nitrogen, a phenol, an aromatic ring and an ethylene hydrocarbon bridg
D. A charged nitrogen, a phenol, an aromatic ring and an *N*-methyl substituent

26. What sort of receptor is the μ receptor?

A. Calcium ion channel
B. Gi-protein linked receptor
C. Potassium ion channel
D. Gs-protein linked recepto

27. Which of the following receptors is considered to be the **safest** when activated?

A. μ
B. κ
C. δ
D. σ

28. Which of the following describes the properties of nalorphine?

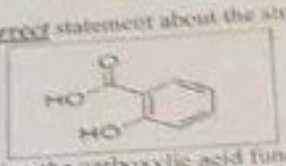
A. It is an agonist at the μ receptor and the κ receptor.
B. It is an antagonist at the μ receptor and the κ receptor.
C. It is an antagonist at the μ receptor and an agonist at the κ recepto
D. It is an antagonist at the κ receptor and an agonist at the μ recept

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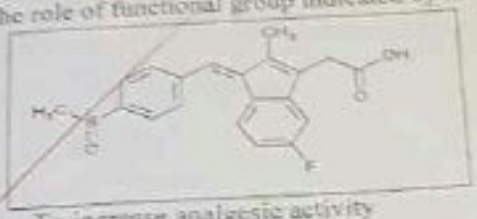
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37. Identify the correct statement about the structure activity relationship of the following structure



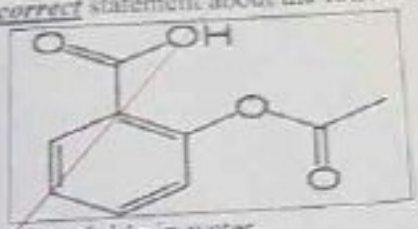
A. Changing the carboxylic acid functional group with amide sustain the anti-inflammatory property of the drug
B. Substitution of halogen atom on the aromatic ring decrease the toxicity
C. Substitution of aromatic ring at C-4 increases the anti-inflammatory property of the drug
D. None of the above

38. What is the role of functional group indicated by arrow in the following structure



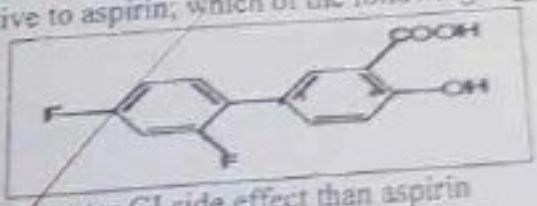
A. To increase analgesic activity
B. To reduce the CNS side effect
C. To increase the GI side effect
D. To increase water solubility

39. Identify the *correct* statement about the following structure



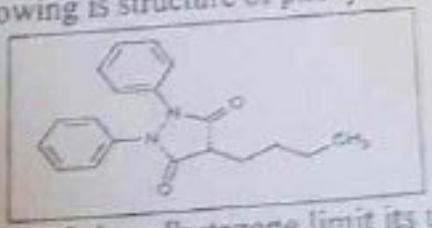
A. Its highly soluble in water
B. Its susceptible for esterase enzyme
C. Reversible acetylator of cyclooxygenase enzyme
D. All

40. Relative to aspirin; which of the following is *not true* about the following structure



A. Has greater GI side effect than aspirin
B. Has longer duration of action than aspirin
C. Its absorbed only from intestine
D. B and C

41. The following is structure of phenylbutazone

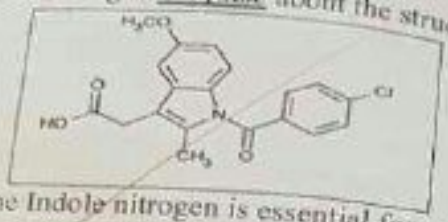


Which toxicity of phenylbutazone limit its use as anti-inflammatory agent

A. Agranulocytosis
B. GI side effect
C. Bone marrow depression
D. All

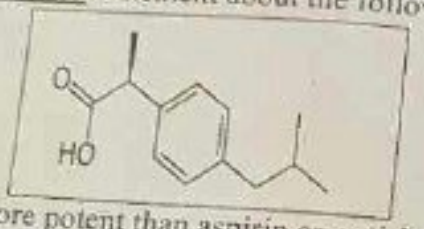
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42. Which of the following is ***not false*** about the structure activity relationship of the



A. The Indole nitrogen is essential for activity
B. Methyl substitution at α-position is less active than aryl substitution
C. Replacing the carboxyl group with other acidic functionality reduce anti-
activity
D. All except C

43. Identify the ***correct*** statement about the following structure



A. More potent than aspirin as anti-inflammatory agent
B. More potent than indomethacin as anti-inflammatory agent
C. The R enantiomer is more active than S enantiomer
D. All
E. All except A

Part II: Essay type questions. (4 pts. each)

1. Briefly discuss the structure activity relationship of opioid analgesics.
2. Barbituric acid lacks any central depressant activity
3. Briefly discuss the structure activity relationship (SAR) of the tricyclic neuroleptics.
4. Compare the activity of these three opiods listed here under.

   Morphine, Codeine, Diacetyl morphine